

Of Errors and Echoes in Tool-Use Agent Credit Assignment

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Abstract

Reinforcement learning improves tool-use agents, yet terminal rewards and step-level feedback can conflate decision errors with their downstream consequences. An initial error may propagate through locally reasonable actions, while later decisions may introduce new errors, miss correction opportunities, or repair earlier mistakes. We introduce Error-Chain-aware Optimization (ECHO), a responsibility-aware credit assignment framework that distinguishes error sources from their subsequent echoes. A frozen judge combines retrospective trajectory analysis with independent policy-visible-prefix assessment to identify newly introduced errors, inherited consequences, missed corrections, and successful corrections. ECHO then allocates shared downstream consequences without double counting and constrains rewards for repairing self-induced errors through source-funded recovery, producing bounded action-level supervision alongside terminal supervision. Experiments on six diverse tool-use benchmarks demonstrate that ECHO achieves strong and consistent gains over baselines across both 4B and 8B scales, ranking first on five benchmarks and second on ACEBench-Agent. These results suggest that decision responsibility provides an effective basis for fine-grained credit assignment in tool-use agents. Our code is available at <https://anonymous.4open.science/r/OctARR-ECHO>.

1 Introduction

Tool-use agents retrieve information and modify environments through sequential decisions (Yao et al., 2022; Schick et al., 2023; Qin et al., 2024). Reliable execution requires tracking state changes and constraints that shape subsequent tool calls (Zhou et al., 2024a; Yao et al., 2024; Lu et al., 2025). Reinforcement learning can improve task performance, but terminal rewards reveal only whether a trajectory succeeds or fails, leaving unresolved which

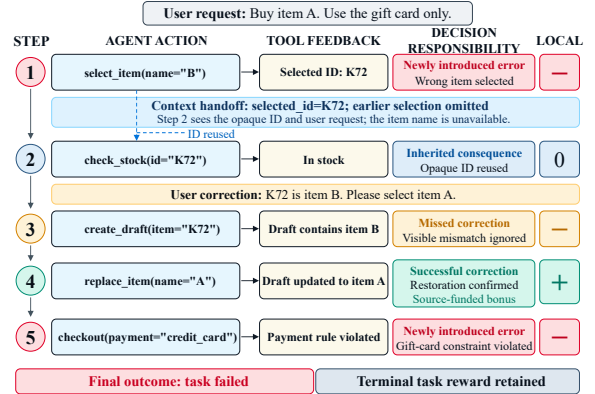


Figure 1: ECHO distinguishes newly introduced errors, inherited consequences, missed corrections, and successful corrections in an illustrative failed trajectory. Local signs denote auxiliary advantage corrections. Verified repairs receive source-funded bonuses, and terminal task rewards are retained.

decisions are responsible for the outcome (Wang et al., 2025b; Qian et al., 2026). Prior work therefore introduces step-level or process-level feedback (Lightman et al., 2024; Wang et al., 2024; Zhang et al., 2025b; Zheng et al., 2025). However, step-level localization alone does not fully resolve credit assignment: the step at which an error manifests is not necessarily the decision that should be held responsible.

Figure 1 illustrates why. The user requests item A and specifies that payment must use the gift card only. The agent first selects the wrong item B and passes its identifier K72 downstream. Step 2 queries inventory using this identifier, even though it refers to the wrong item B. The user later reveals that K72 refers to B rather than A, yet Step 3 still fails to correct the mistake. Step 4 restores the intended item A, while Step 5 independently violates the payment constraint by using a credit card instead of the gift card. The trajectory therefore contains error introduction, propagation, missed correction, and recovery, raising two distinct credit-assignment questions.

First, which decisions should be held responsible? For propagated errors, the key distinction is whether the discrepancy was identifiable when the action was taken. Step 2 (*inventory query with K72*) inherits an erroneous state, but the mismatch is not visible in its local context. Step 3 (*continued use after disclosure*), by contrast, ignores a mismatch that has already become visible and thus constitutes a missed correction. Responsibility must therefore be assessed from decision-time information, rather than from the erroneous state alone or from evidence revealed later.

Second, once responsible decisions are identified, how should credit be allocated? Even if step-level supervision correctly identifies the responsible steps, assigning rewards to them independently cannot properly account for shared consequences or subsequent recovery. Step 1 (*wrong item selection*) and Step 3 (*missed correction*) both contribute to the same wrong-item consequence, so assigning each the full penalty would double-count the same impairment. Step 4 (*item repair*) introduces a different challenge: recovery may deserve positive credit, but not in a way that rewards the policy for first creating its own repair opportunity. Step 5 (*payment violation*), meanwhile, is an independent defect. Credit assignment must therefore account for shared consequences and recovery, rather than scoring each responsible step independently.

We introduce **ECHO** (**E**rror-**C**hain-aware **O**ptimization), which addresses these two problems jointly by assigning credit over error chains rather than treating steps independently. ECHO distinguishes four responsibility events: ① newly introduced errors, ② inherited consequences, ③ missed corrections, and ④ successful corrections. ECHO first performs responsibility identification. A frozen trajectory judge uses the complete trajectory to propose candidate events and establish propagation and restoration evidence, while the local responsibility of each selected action is independently assessed using only its policy-visible prefix. In Figure 1, Step 2 (*inventory query with K72*) is treated as an inherited consequence, whereas Step 3 (*continued use after disclosure*) is identified as a missed correction. Later observations can therefore verify propagation and restoration without leaking future information into the responsibility judgment of an earlier decision.

ECHO then performs credit allocation. When multiple responsible decisions contribute to the same downstream consequence, shared-

consequence accounting counts that consequence once and allocates it across supported sources of responsibility. For repairs of self-induced errors, source-funded recovery constrains positive correction credit by the budget supplied by the corresponding source error. Accordingly, Step 1 (*wrong item selection*) and Step 3 (*missed correction*) are not both charged the full wrong-item consequence, while the repair credit at Step 4 (*item repair*) is bounded by its originating error. These responsibility signals are converted into bounded action-level advantage corrections and optimized alongside the original terminal supervision.

We evaluate ECHO at 4B and 8B scales across six tool-use benchmarks. At both scales, ECHO ranks first among the compared models on five of six benchmarks. Further ablation and frozen-judge analyses examine the roles of responsibility identification, credit allocation, and judge quality.

Our contributions are threefold:

- We formulate local supervision through decision responsibility, separating error sources, inherited consequences, and correction behavior.
- We develop ECHO to connect independent prefix assessment with bounded consequence and recovery accounting for fine-grained policy optimization.
- We establish cross-scale and cross-benchmark evidence and examine the effects of error selection and frozen-judge choice through targeted comparisons.

2 Related Work

Tool-Use Agent Learning. Reinforcement learning has substantially improved tool-use agents through reward design, multi-turn rollouts, and simulated users (Qian et al., 2026; Feng et al., 2026a; Zhao et al., 2025; Dong et al., 2026; Zhou et al., 2024b; Wen et al., 2024; Luo et al., 2025), as well as through scaled training environments (Song et al., 2026; Wang et al., 2026b, 2025a; Li et al., 2025). These methods primarily optimize trajectory-level outcomes or scale the environments used for training. ECHO is complementary, addressing how supervision is distributed across individual agent decisions when errors propagate, become observable, or are later corrected.

Process Supervision and LLM Critics. Step-level feedback provides denser supervision than terminal rewards (Lightman et al., 2024; Wang et al., 2024; Zhang et al., 2025b; Zheng et al., 2025), and

reward models and LLM critics extend this idea to search and tool-use agents (Xia et al., 2025; Li et al., 2026b; Xi et al., 2026; Tan et al., 2025, 2026; Zhou et al., 2025; Zhang et al., 2026). Such signals localize problematic steps, but localization alone does not establish decision responsibility: an observed error may be inherited from an earlier decision, and judging a decision with later evidence can penalize it for information unavailable when it was made. ECHO therefore uses the full trajectory only to propose candidate events and verify propagation or restoration, assessing each decision independently from its policy-visible prefix.

Credit Assignment and Error Attribution. Beyond identifying responsible decisions, a complementary question is how delayed outcomes should be allocated among them. Reward redistribution and recent agent methods estimate step or turn contributions from delayed returns, trajectory structure, counterfactual rollouts, or turn-level value estimates (Arjona-Medina et al., 2019; Meulemans et al., 2023; Zhang et al., 2023; Feng et al., 2026b; Li et al., 2026a; Wang et al., 2026a; Gandhi et al., 2026; Meng and Chen, 2026; Tao et al., 2026; Liao et al., 2026; Zhan et al., 2026), while failure-attribution studies identify root-cause errors for post-hoc diagnosis (Zhang et al., 2025a; Zhu et al., 2025). Conceptually related to structural accounts of responsibility (Chockler and Halpern, 2004), ECHO explicitly separates responsibility identification from credit allocation: shared-consequence accounting counts each consequence once across its responsible decisions, and source-funded recovery bounds self-induced repair credit by the budget of its originating error.

3 Method

ECHO converts decision-responsibility evidence into bounded, signed action-level advantage corrections, as illustrated in Figure 2. Independent local assessment, shared-consequence accounting, and source-funded recovery determine the corrections before policy optimization.

3.1 Interaction Traces and Responsibility Events

A policy π_θ completes task x through interactions with tools and users. Trajectory i comprises actions $a_{i,t}$, subsequent observations $o_{i,t}$, and terminal reward R_i . The pre-action context $h_{i,t}$ records the information actually visible to the policy after con-

text truncation or memory retrieval. The execution harness preserves these contexts, tool results, available state changes, and policy-token spans. Local responsibility is assessed against this recorded policy-visible information.

Responsibility is assessed jointly by action and predicate. An event records

$$e = \langle t, f, d, z, \mathcal{W}_e^{\text{pre}}, \mathcal{W}_e^{\text{post}}, u_e \rangle, \quad (1)$$

where f is a versioned obligation or state predicate, d its deviation origin, and z the responsibility type. Evidence pointers $\mathcal{W}_e^{\text{pre}}$ and $\mathcal{W}_e^{\text{post}}$ identify the supporting messages or structured fields. For a missed correction, witness u_e specifies a feasible response. An action can repair one predicate while violating another, producing separate events whose training coefficients are subsequently aggregated.

3.2 Evidence-Grounded Responsibility Assessment

Responsibility types. A *newly introduced error* is a distinct decision defect supported by a policy-visible fact or constraint. A *missed correction* combines an existing deviation, visible discrepancy evidence, an available permitted response, and failure to reasonably address the deviation. Feasibility is assessed against the specific local issue. An *inherited consequence* has explicit provenance from an earlier deviation, with neither a supported new defect nor a missed correction for the relevant predicate. Ambiguous classifications trigger abstention. A *successful correction* combines a legitimate response to a visible deviation with execution evidence confirming restoration of the predicate or a required precondition. Unverified repair claims receive zero positive weight.

Ordinary correct actions retain terminal supervision with zero auxiliary correction. Responsibility for external failures concerns the agent’s response to the observed incident. Events sharing an action, predicate, origin, and type are merged; competing labels resolve to one supported classification or abstention.

Independent prefix assessment. A frozen trajectory judge first proposes event locations, propagation links, and restoration evidence from the complete trajectory. At most K selected actions receive independent assessments using $h_{i,t}$, $a_{i,t}$, and the tool contracts visible at that time. The fixed assessment prompt identifies supported local defects or legitimate repair intentions. Its input excludes

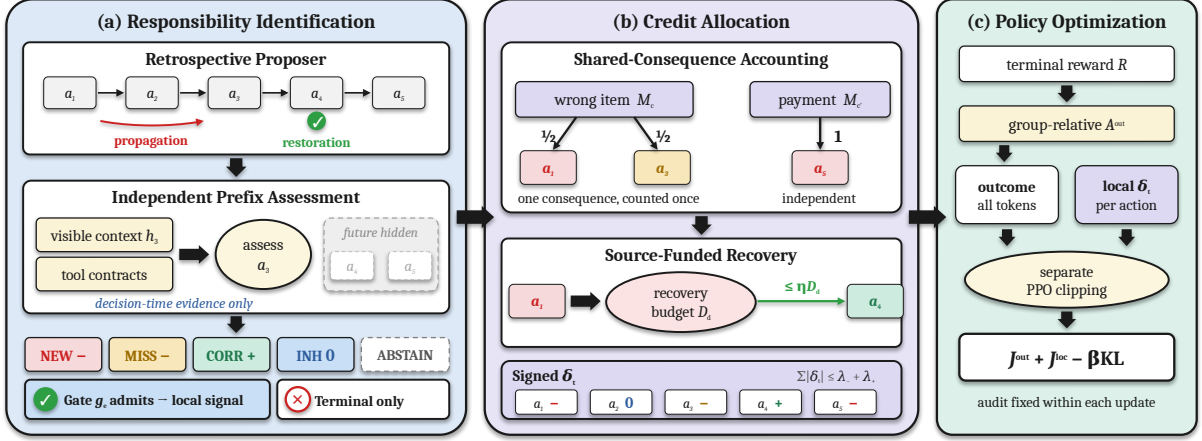


Figure 2: Overview of ECHO. Independent prefix assessment establishes local responsibility. Shared-consequence accounting and source-funded recovery produce bounded action-level corrections for a separately clipped, length-balanced local objective alongside terminal supervision. Signs denote auxiliary advantage corrections.

retrospective proposals, hidden state, and future observations, preserving the policy’s decision-time information boundary. Successful corrections additionally require retrospective execution confirmation.

Under deterministic inference with fixed decoding, a queried action’s local assessment is invariant to its suffix. Retrospective evidence still determines candidate selection and impact weights. The assessment budget reserves six slots for proposed actions and two for non-proposed sentinel actions, with selection rules in Appendix A.

Selective supervision. Events undergo evidence-pointer, visibility, temporal-order, provenance, and nonempty-span checks. A binary gate g_e admits evidence strata satisfying a calibrated semantic-error bound estimated from independent task-level annotations. Events with insufficient evidence, ambiguous origins, unsupported restoration, or failed checks abstain from local supervision. Appendix B defines the gate and its stratum-level guarantee under the stated sampling assumptions.

3.3 Shared Consequences and Propagation Weighting

Constraint-impairment proxy. A constraint-impairment proxy estimates attributable loss from evidenced decision defects and observed impairment of task conditions. Before inspecting outcomes, a fixed task adapter extracts nonredundant requirement predicates $\mathcal{F}_x$ from public instructions and tool contracts. Predicate masses ω_f sum to one and are uniform by default. Predicate versions share their original obligation’s mass, which

is partitioned among primitive subfacts. Obligations outside this inventory contribute through the direct-defect term.

Repeated manifestations of a predicate’s impairment form one *consequence atom* c , with

$$M_c = \omega_{f(c)} s_c, \quad \sum_c M_c \leq 1. \quad (2)$$

Severity s_c is zero for unconfirmed impairment, $1/2$ for reversible impairment, and one for terminal or irreversible impairment. Recovery preserves the recorded source defect and maximum authenticated severity.

Divisible ownership. Let P_c contain the responsibility slots explicitly linked to consequence c , including uncertified slots. Reserve fraction ζ_c is zero for complete recorded provenance, $1/2$ for unresolved additional sources, and one for unsupported or conflicting ownership. The allocated share is

$$a_{ec} = \begin{cases} \frac{1 - \zeta_c}{|P_c|}, & e \in P_c, |P_c| > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Equal division provides an observational accounting convention, with rejected shares remaining unallocated. A source error and a subsequent missed correction receive distinct shares of their common consequence. Validated propagation links identify explicit reuse of corrupted values or invalidated conditions.

Event weights. For a negative event, define

$$\widehat{\ell}_e^- = \kappa d_e + (1 - \kappa) \sum_c M_c a_{ec}, \quad (4)$$

where d_e indicates a certified distinct decision defect and $0 < \kappa \leq 1$. The direct term counts distinct certified decision defects, including repaired errors and defects with unresolved downstream contributions. Repeated observations remain assigned to their existing event.

For a successful correction r , first-restoration shares z_{rc} satisfy $\sum_r z_{rc} \leq 1$, giving

$$\hat{\ell}_r^+ = \kappa + (1 - \kappa) \sum_c M_c z_{rc}. \quad (5)$$

Each verified restoration contributes its allocated share and direct term once. Let $\mathcal{D}_e$ contain distinct requirement accounts reached through validated propagation links. For either event direction,

$$p_e = \sum_{f \in \mathcal{D}_e} \omega_f, \quad v_e = g_e \hat{\ell}_e [1 + \alpha p_e], \quad (6)$$

where $0 \leq \alpha \leq 1$ and $0 \leq p_e \leq 1$. For corrections, $\mathcal{D}_e$ is restricted to the repaired origin's footprint. Canonical requirement accounts preserve propagation scope under duplicated log entries for fixed validated records. The multiplier bounds amplification of shared consequence loss by $1 + \alpha$, with direct-defect terms accounted for separately.

3.4 Source-Funded Recovery Budgets

Let $\mathcal{E}_i^-$ contain admitted newly introduced errors and missed corrections, and $\mathcal{E}_i^+$ contain eligible execution-verified corrections. Inherited consequences and abstained events receive zero auxiliary budget while retaining terminal supervision. For valid policy spans, provisional directional budgets are

$$b_e^s = \lambda_s \frac{v_e}{\max\{1, \sum_{u \in \mathcal{E}_i^s} v_u\}}, \quad s \in \{-, +\}. \quad (7)$$

Each direction is capped at λ_s , while weight sums below one retain their unnormalized scale.

To bound credit for self-created repair opportunities, ECHO classifies deviation origins as self-induced, authenticated external, or unknown. For a self-induced origin d , the source-funded recovery principal is

$$D_d = \sum_{e: z_e = \text{NEW}, d(e) = d} b_e^-. \quad (8)$$

Newly introduced errors fund this principal, with each source charge assigned to one origin. Thus,

the item-selection error in Figure 1 funds its verified repair, while the payment violation carries separate responsibility. Let $\mathcal{R}_d$ contain eligible restorations and $B_d^+ = \sum_{r \in \mathcal{R}_d} b_r^+$. For $B_d^+ > 0$, set $\gamma_d = \min\{1, \eta D_d / B_d^+\}$, and otherwise set $\gamma_d = 0$. Paid correction coefficients are

$$\tilde{b}_r^+ = \begin{cases} \gamma_d b_r^+, & d \text{ self-induced,} \\ b_r^+, & d \text{ authenticated external,} \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Positive normalization covers authenticated external recoveries and self-induced recoveries with a charged source. Partial repairs share their origin's principal; unused credit is discarded. Each authenticated primitive external incident permits one recovery entitlement.

Aggregating events on each action yields

$$\delta_{i,t} = \sum_{r \in \mathcal{E}_{i,t}^+} \tilde{b}_r^+ - \sum_{e \in \mathcal{E}_{i,t}^-} b_e^-. \quad (10)$$

The aggregated corrections satisfy $\sum_t |\delta_{i,t}| \leq \lambda_- + \lambda_+$ and enter the local objective unchanged. Positive and negative event records remain available when their action-level coefficients cancel.

3.5 Length-Balanced Policy Optimization

Let $q_{i,k}$ indicate a loss-eligible policy token and $m_{i,t,k}$ its membership in action t . The masks satisfy $m_{i,t,k} \leq q_{i,k}$ and $\sum_t m_{i,t,k} \leq 1$, restricting local loss to nonoverlapping, eligible policy spans. Set $L_i = \sum_k q_{i,k}$ and $B_{i,t} = \sum_k m_{i,t,k}$. Zero-length actions abstain; empty policy sequences contribute zero policy loss.

For trajectories sampled from the same task and initial conditions, group-relative outcome advantages are (Shao et al., 2024)

$$A_i^{\text{out}} = \frac{R_i - \mu_x}{\sigma_x + \varepsilon}, \quad (11)$$

using all group rewards and their population standard deviation. Equal-reward groups retain zero outcome advantage but can receive supported local corrections.

Let $\rho_{i,k}$ be the current-to-behavior token likelihood ratio under the stored policy context. Using the proximal policy optimization clipped surrogate (Schulman et al., 2017), define

$$\phi_\epsilon(\rho, A) = \min\{\rho A, \text{clip}(\rho, 1 - \epsilon, 1 + \epsilon)A\}. \quad (12)$$

Model		BFCL-v3 MT Overall	BFCL-v4 Overall	τ -bench Overall	τ^2 -Bench Overall	ToolSandbox Overall	ACEBench- Agent Overall
4B	Qwen3	25.38	38.61	33.44	15.83	52.89	55.28
	EnvScaler	37.25	<u>52.92</u>	48.78	<u>33.05</u>	62.25	63.33
	Arctic-AWM	<u>38.89</u>	49.55	26.58	22.57	<u>68.24</u>	60.58
	ECHO (ours)	41.24	53.71	50.77	35.81	76.36	<u>62.46</u>
8B	Qwen3	28.88	41.58	36.65	26.44	59.74	60.00
	EnvScaler	41.38	54.20	44.22	35.10	<u>68.85</u>	58.33
	Arctic-AWM	<u>43.70</u>	52.50	37.64	33.45	60.44	54.93
	Klear-AgentForge	37.84	47.88	<u>49.10</u>	35.91	66.06	60.70
	MUA-RL	14.63	26.84	43.00	30.20	50.00	53.30
	Simia-RL	41.64	<u>55.72</u>	44.27	<u>40.16</u>	62.25	58.33
	ECHO (ours)	49.17	56.91	50.59	40.56	74.42	<u>60.08</u>

Table 1: Overall results on six tool-use benchmarks at the 4B and 8B scales, spanning multi-turn function calling, conversational task completion, and stateful tool execution. For BFCL-v3, we report the overall score on the Multi-Turn (MT) subset. For ACEBench, we report results on the Agent subset. Scores are percentages. **Bold** and underlined values mark the best and second-best results within each scale.

ECHO maximizes

$$J_i^{\text{out}} = \frac{1}{L_i} \sum_k q_{i,k} \phi_\epsilon(\rho_{i,k}, A_i^{\text{out}}), \quad (13)$$

$$J_i^{\text{loc}} = \sum_{t: B_{i,t} > 0} \frac{1}{B_{i,t}} \sum_k m_{i,t,k} \phi_\epsilon(\rho_{i,k}, \delta_{i,t}), \quad (14)$$

$$J_{\text{ECHO}} = \frac{1}{N} \sum_i [J_i^{\text{out}} + J_i^{\text{loc}} - \beta K_i], \quad (15)$$

where K_i is the trajectory-mean reference-policy KL penalty and N counts contributing trajectories. Judge outputs, rewards, budgets, and advantages are fixed during each update. The local branch averages over tokens within each action, sums across actions, and averages across trajectories.

Action-level averaging preserves local coefficients that could otherwise be attenuated by magnitude clipping of length-rescaled token advantages. Optimization combines uncapped local advantage magnitudes with likelihood-ratio clipping, KL control, and gradient-norm clipping. At the sampling policy, the split objective has the same gradient as the uncapped token advantage

$$A_i^{\text{out}} + L_i \sum_{t: B_{i,t} > 0} \frac{\delta_{i,t}}{B_{i,t}} m_{i,t,k}. \quad (16)$$

Once ratio clipping activates, opposite-sign advantages can enter different branches, producing distinct split and combined objectives.

For a fixed audited trajectory whose positive events are exclusively self-funded, define $N_i^- = \sum_t \max\{-\delta_{i,t}, 0\}$. The escrow constraint yields

$$J_i^{\text{loc}} \leq [(1 + \epsilon)\eta - (1 - \epsilon)]N_i^- \leq 0 \quad (17)$$

when $\eta \leq (1 - \epsilon)/(1 + \epsilon)$ and $0 < \epsilon < 1$. This fixed-batch bound concerns the local surrogate; its behavioral implications depend on attribution accuracy and shared policy parameters. §C derives the bounds, and §E details optimization and computational cost.

4 Experiments

We evaluate ECHO along three dimensions. First, we assess its overall performance across six tool-use benchmarks at the 4B and 8B scales (§4.2). Second, we analyze the contributions of its design choices through component and reward ablations (§4.3 and §4.4). Finally, we examine how frozen-judge choice relates to responsibility identification and downstream policy performance (§4.5).

4.1 Setup

Benchmarks and metrics. We evaluate ECHO at the 4B and 8B scales on BFCL-v3 Multi-Turn, BFCL-v4, τ -bench, τ^2 -Bench, ToolSandbox, and ACEBench-Agent. BFCL evaluates function calling, with v4 additionally covering web search and memory use (Patil et al., 2025; Mao et al., 2025). We evaluate both τ -bench domains, retail and airline (Yao et al., 2024), and all three τ^2 -Bench domains, retail, airline, and telecom (Barres et al., 2025). These conversational benchmarks test policy-constrained task completion, with telecom additionally requiring coordinated agent-user actions. ToolSandbox evaluates intermediate and final state milestones (Lu et al., 2025), while ACEBench-Agent covers multi-step and multi-turn tool use in English and Chinese (Chen et al., 2025).

Model		τ	τ	τ	τ^2	τ^2	τ^2	τ^2
		Retail	Airline	Overall	Retail	Airline	Telecom	Overall
4B	Qwen3	40.87	26.00	33.44	19.96	21.00	9.43	15.83
	EnvScaler	49.57	48.00	48.78	50.00	36.00	13.16	33.05
	Arctic-AWM	30.15	23.00	26.58	30.26	19.00	16.45	22.57
	ECHO (ours)	52.03	49.50	50.77	54.09	36.00	17.33	35.81
8B	Qwen3	45.79	27.50	36.65	34.43	26.50	18.42	26.44
	EnvScaler	50.43	38.00	44.22	53.51	36.00	15.79	35.10
	Arctic-AWM	39.88	35.40	37.64	41.23	38.50	23.47	33.45
	Klear-AgentForge	56.70	41.50	49.10	49.42	36.67	21.64	35.91
	MUA-RL	56.50	29.50	43.00	49.80	19.00	21.80	30.20
	Simia-RL	51.82	36.71	44.27	52.90	49.00	18.58	40.16
	ECHO (ours)	58.17	43.00	50.59	54.63	42.10	24.95	40.56

Table 2: Domain-level results on τ -bench and τ^2 -Bench at the 4B and 8B scales. All scores are reported as percentages. **Bold** and underlined values mark the best and second-best results within each model scale.

Model		Base	Missing Function	Missing Parameter	Long Context	Overall
4B	Qwen3	32.00	20.00	24.00	25.50	25.38
	EnvScaler	46.50	38.50	28.50	35.50	37.25
	Arctic-AWM	48.55	36.50	34.22	36.30	38.89
	ECHO (ours)	50.50	46.58	32.59	35.27	41.24
8B	Qwen3	32.00	33.50	22.00	28.00	28.88
	EnvScaler	50.50	41.50	29.50	44.00	41.38
	Arctic-AWM	53.40	45.00	32.40	44.00	43.70
	Klear-AgentForge	45.86	39.00	34.50	32.00	37.84
	MUA-RL	21.00	11.50	15.00	11.00	14.63
	Simia-RL	54.50	36.50	37.54	38.00	41.64
	ECHO (ours)	59.50	51.50	40.68	45.00	49.17

Table 3: BFCL-v3 Multi-Turn results across Base, Missing Function, Missing Parameter, and Long Context conditions at the 4B and 8B scales. Overall is the unweighted mean across the four conditions. All scores are reported as percentages. **Bold** and underlined values mark the best and second-best results within each model scale.

Baselines. Qwen3-4B and Qwen3-8B serve as the base policies at 4B and 8B, respectively (Yang et al., 2025). At 4B, we compare against EnvScaler (Song et al., 2026) and Arctic-AWM (Wang et al., 2026b). At 8B, we additionally compare against Klear-AgentForge (Wang et al., 2025a), MUA-RL (Zhao et al., 2025), and Simia-RL (Li et al., 2025).

Training and configuration. ECHO is trained from Qwen3-4B and Qwen3-8B on the τ^2 -Bench training split. Training details and hyperparameters are given in §D. The frozen-judge protocol and reliability gating are described in §A and §B.

4.2 Main Results

Overall performance. ECHO achieves the best result on five of six benchmarks at both model scales. At 4B, it outperforms the strongest same-scale baseline by 2.35 points on BFCL-v3 Multi-Turn, 0.79 on BFCL-v4, 1.99 on τ -bench, 2.76 on τ^2 -Bench, and 8.12 on ToolSandbox. At 8B, the corresponding gains are 5.47, 1.19, 1.49, 0.40, and 5.57 points. ACEBench-Agent is the only benchmark on which ECHO does not lead, trailing

the best baseline by 0.87 points at 4B and 0.62 at 8B. The ranking pattern is consistent across scales: ECHO leads on the same five benchmarks at 4B and 8B, with the largest margins on BFCL-v3 Multi-Turn and ToolSandbox.

Domain-level results. ECHO leads or ties the strongest baseline on all five domain-level comparisons at 4B. At 8B, it leads τ -bench Retail and Airline by 1.47 and 1.50 points, and τ^2 -Bench Retail and Telecom by 1.12 and 1.48 points, respectively. The main exception is τ^2 -Bench Airline, where ECHO scores 42.10 versus 49.00 for Simia-RL. Despite this 6.90-point deficit, ECHO achieves the highest τ^2 -Bench overall score, 40.56 versus 40.16 for Simia-RL, as its gains on Retail and Telecom outweigh the Airline deficit.

BFCL-v3 condition-level results. At 4B, ECHO leads on Base and Missing Function by 1.95 and 8.08 points, respectively, but trails the strongest baseline by 1.63 points on Missing Parameter and 1.03 on Long Context. At 8B, ECHO leads all four conditions, with gains of 5.00 points on Base, 6.50

on Missing Function, 3.14 on Missing Parameter, and 1.00 on Long Context. This advantage extends to all four conditions at 8B, whereas at 4B it is concentrated in Base and Missing Function.

4.3 Component Ablation

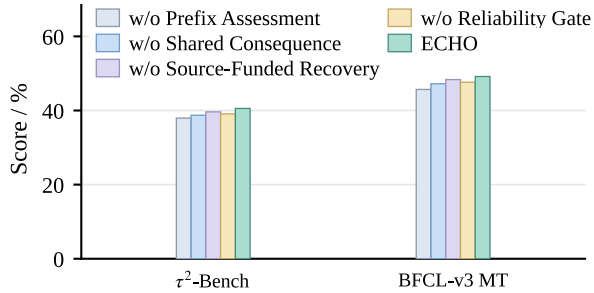


Figure 3: Ablation of ECHO components on τ^2 -Bench and BFCL-v3 Multi-Turn at the 8B scale.

Figure 3 presents one-component-at-a-time ablations of ECHO. w/o Prefix Assessment omits independent prefix assessment; w/o Shared Consequence assigns full consequence mass to each event; w/o Source-Funded Recovery removes source-funded recovery constraints; and w/o Reliability Gate disables calibrated semantic filtering. All four ablations reduce performance on both benchmarks, with a consistent ordering of drops. Removing prefix assessment causes the largest degradation, suggesting a central role for decision-time responsibility, whereas removing source-funded recovery has the smallest effect.

4.4 Reward Ablation

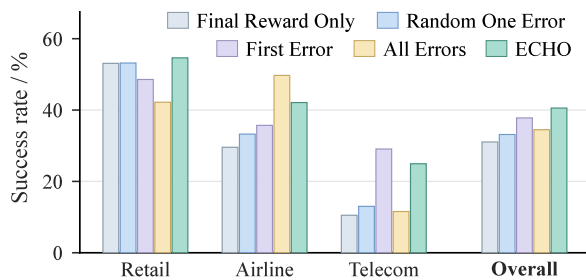


Figure 4: Reward ablation on τ^2 -Bench at the 8B scale. Each group compares five supervision configurations, and Overall denotes the mean across the three domains.

At the 8B scale, we compare ECHO with Final Reward Only, Random One Error, First Error, and All Errors. Final Reward Only retains terminal supervision, while the three heuristics penalize one random error, the first error, or all identified errors, respectively. ECHO exceeds Final Reward Only by 9.51 points and First Error, the strongest heuristic, by 2.77 points overall. Although All Errors and First Error lead Airline and Telecom, respectively,

ECHO achieves the highest Retail score and the highest overall score. These results suggest that ECHO benefits not simply from additional local supervision, but from responsibility-aware credit assignment across errors and corrections. §F reports the same comparison on BFCL-v3 Multi-Turn.

4.5 Effect of the Frozen Judge

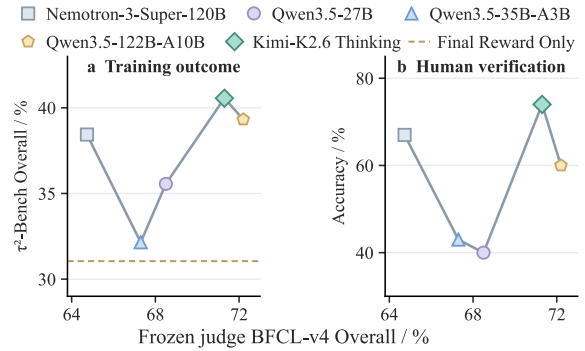


Figure 5: Frozen-judge comparison at the 8B scale. Left: downstream ECHO performance on τ^2 -Bench. Right: responsibility identification accuracy on 100 manually verified samples before reliability gating.

Frozen-judge comparisons examine whether general tool-use capability predicts effectiveness as a source of local supervision. Figure 5 compares five judges by BFCL-v4 score, downstream ECHO-8B performance, and responsibility identification accuracy on 100 manually verified samples. Nemotron-3-Super-120B has the lowest BFCL-v4 score, yet yields stronger downstream ECHO performance than both smaller Qwen judges. Kimi-K2.6 Thinking produces the strongest ECHO policy, exceeding Qwen3.5-122B-A10B by 1.24 percentage points despite the latter’s higher BFCL-v4 score. These reversals suggest that general tool-use capability is an incomplete proxy for supervision effectiveness.

5 Conclusion

We presented ECHO, a responsibility-aware credit assignment framework for tool-use agents that distinguishes newly introduced errors, inherited consequences, missed corrections, and successful corrections, and converts them into bounded action-level supervision. This design addresses three limitations of conventional terminal or step-level feedback: (i) conflating error manifestations with responsible decisions, (ii) double-counting shared downstream consequences, and (iii) over-crediting recovery from self-induced errors. Across six tool-use benchmarks and two model scales, ECHO achieves strong and broadly consistent improve-

ments over the compared baselines. Together, these results support decision responsibility as a useful perspective for fine-grained credit assignment across complex tool-use trajectories.

Limitations

This work focuses specifically on responsibility-aware credit assignment for tool-use agents. (i) ECHO is designed for training-time credit assignment rather than inference-time intervention. Techniques such as reflection, replanning, and online verification are complementary to the proposed framework and are not considered here. (ii) Our analysis centers on responsibility attribution within agent–tool interaction trajectories. Broader system-level considerations, including tool construction, multi-agent coordination, and interface design, fall outside the scope of the proposed framework.

Ethical Considerations

All experiments in this study were conducted using publicly available datasets and models licensed for research use. To the best of our knowledge, we do not identify any method-specific ethical concerns beyond those generally associated with tool-use agents and automated model evaluation.

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A Audit Protocol and Evidence Handling

A.1 Candidate Selection and Independent Requests

The full-trajectory proposer returns at most sixteen event hypotheses and explicitly marks omitted records. Each hypothesis identifies its action, policy span, versioned predicate, deviation origin, evidence pointers, and proposed responsibility type. Impact records specify propagation links, severity, first-restoration witnesses, and authenticated origins. Validated fields determine training weights through the scoring rules in Section 3.3.

The prefix-review budget covers $K = 8$ distinct actions. Up to six proposed actions are selected chronologically; two non-proposed sentinel actions follow a fixed-seed hash ordering. This ordering also fills unused proposal slots until the budget or available actions are exhausted. Each request returns at most four predicate-level records, with omitted records assigned abstention.

Each prefix request contains the recorded policy-visible context, current action, and available tool

contracts. Retrospective labels, selected predicates, rationales, severity, suffixes, and hidden state are excluded from these inputs. Batched requests retain isolated attention contexts, and both inference roles use the same frozen checkpoint with fixed prompts and decoding. Prefix assessment returns supported defects, legitimate repair intentions, defect-free assessments, or abstention. The assembler combines an accepted repair intention with execution confirmation to establish a successful correction.

Sentinel-discovered defects contribute through the direct term when their prefix evidence passes validation. Unsupported impact and propagation terms receive zero weight, and sentinel repairs require authenticated restoration evidence for positive supervision.

A.2 Responsibility Boundaries

Inherited inputs and current obligations. The classification of a reused product identifier depends on the discrepancy evidence visible when it is used. A visible mismatch and an available resolving lookup can establish a missed correction when the agent continues committing to the wrong product. An apparently valid opaque identifier can carry an inherited consequence when its defect becomes observable later. Ambiguous cases receive local abstention, preserving the distinction between available evidence and retrospectively established error.

Feasible responses. A missed-correction witness specifies a concrete tool or response, known arguments, availability under the visible contract, and the discrepancy addressed. Clarification qualifies when the user can disambiguate exposed alternatives; stopping qualifies when permitted and sufficient to avoid the identified impermissible operation. A legitimate repair attempt retains its prefix-grounded assessment even if an unpredictable outage later prevents restoration.

Mixed actions and verified restoration. An action that restores one field while violating another produces distinct positive and negative events. Transaction identifiers or field-level provenance link verified restoration to the responsible action. Ambiguous restoration attribution triggers abstention from positive supervision. A verified correction remains valid despite a separate terminal failure, with auxiliary eligibility determined by its origin and available funding.

Visibility and external origins. Decision-time visibility follows delivery timestamps and actual context inclusion, placing synchronous tool output after its generating action. Hidden snapshots support retrospective impairment assessment, while correction opportunities require policy-visible discrepancy evidence. Initial-state records authenticate pre-existing defects, and environment records authenticate subsequent external incidents. Omitted sources and unsupported policy claims leave origins unknown. A new policy error that reopens a repaired predicate creates a separate self-induced episode and source-funded account.

B Calibration of Selective Supervision

Prompt development, gate calibration, hyperparameter selection, and final evaluation use disjoint task partitions. A fixed selection rule samples one eligible event per independent task instance within each predeclared evidence stratum. Events and rollouts sharing a task form one sampling cluster. Two annotators assess prefix-grounded responsibility, followed by a separate retrospective review of impact, provenance, and restoration. Adjudication is blinded to training coefficients, and correctness requires validity of every training-relevant semantic field.

For stratum b , let k_b of n_b independent calibration units contain an incorrect assertion. With B_{cal} predeclared strata and simultaneous failure level δ_{cal} , define

$$U_b = \text{Beta}^{-1} \left[1 - \frac{\delta_{\text{cal}}}{B_{\text{cal}}}; k_b + 1, n_b - k_b \right] \quad (18)$$

for $0 \leq k_b < n_b$, and set $U_b = 1$ if $k_b = n_b$ or $n_b = 0$. The inverse beta distribution yields a one-sided exact binomial upper confidence limit. Combining it with structural validity indicator V_e gives

$$g_e = V_e \mathbf{1}\{n_{b(e)} \geq n_{\min}\} \mathbf{1}\{U_{b(e)} \leq \epsilon_J\}. \quad (19)$$

Independent sampling from the same stratum distributions gives simultaneous upper coverage of at least $1 - \delta_{\text{cal}}$ through a union bound. This guarantee concerns stratum-level semantic-error rates under the calibration distribution. Individual-event validity, impact magnitude, and reliability under policy shift remain separate assessment targets. Post hoc stratum selection uses an independent selection split to preserve the calibration protocol.

The calibration design crosses three supervised responsibility types with explicit or semantic evidence to form six strata. Thresholds are $\delta_{\text{cal}} = 0.05$, $\epsilon_J = 0.10$, and $n_{\min} = 50$; insufficiently sampled or unreliable strata abstain. Precision and coverage jointly characterize selective supervision, while monitoring newly annotated training traces supports drift detection and recalibration on fresh development data.

C Derivations and Guarantee Boundaries

C.1 Shared-Consequence and Budget Bounds

Equation 3 gives $\sum_e a_{ec} = 1 - \zeta_c \leq 1$ for nonempty P_c , and zero otherwise. Let v_e^{shared} denote the shared-consequence component of a negative event's weight. Since $g_e \leq 1$ and $p_e \leq 1$,

$$\begin{aligned} \sum_e v_e^{\text{shared}} &= (1 - \kappa) \sum_c M_c \sum_e a_{ec} g_e [1 + \alpha p_e] \\ &\leq (1 - \kappa)(1 + \alpha) \sum_c M_c. \end{aligned} \quad (20)$$

The shared-consequence bound is independent of the number of implicated events; direct-defect terms separately supervise certified decisions. Duplicating observations preserves the accounting when validated predicates, event keys, provenance links, and first-restoration sets are fixed. The invariance therefore concerns validated records, with proposer sensitivity to redundant text requiring empirical assessment.

Equation 7 caps each directional total at λ_s . Source-funded escrow reduces or preserves positive budgets, giving

$$\sum_t |\delta_{i,t}| \leq \sum_e b_e^- + \sum_r \tilde{b}_r^+ \leq \lambda_- + \lambda_+. \quad (21)$$

The resulting bound limits total auxiliary budget, while additional responsibility events can reduce individual normalized shares.

C.2 First-Order Equivalence

At the behavior policy, likelihood ratios equal one and lie inside the clipping interval. Writing the token score gradient as $s_{i,k} = \nabla_{\theta} \log \pi_{\theta}[y_{i,k} \mid h_{i,k}]$, the uncapped additive-token surrogate gives

$$\begin{aligned} \nabla J_i^{\text{add}} &= \frac{A_i^{\text{out}}}{L_i} \sum_k q_{i,k} s_{i,k} \\ &\quad + \sum_{t: B_{i,t} > 0} \frac{\delta_{i,t}}{B_{i,t}} \sum_k m_{i,t,k} s_{i,k} \\ &= \nabla [J_i^{\text{out}} + J_i^{\text{loc}}]. \end{aligned} \quad (22)$$

Objective values also agree whenever all relevant ratios lie strictly inside the clipping interval. Outside it, opposite-sign advantages can activate different branches, yielding distinct objectives with an identical KL term.

If each token score gradient has norm at most G_0 at the behavior policy, then

$$\|\nabla J_i^{\text{loc}}\| \leq G_0 \sum_t |\delta_{i,t}| \leq G_0[\lambda_- + \lambda_+]. \quad (23)$$

This gradient bound holds under the score-gradient assumption at the sampling policy. Extending it across tokenizations or away from that policy requires additional assumptions.

C.3 Escrow and the Clipped Local Surrogate

For a fixed audited trajectory with exclusively self-funded positive events, define gross budgets $P_0 = \sum_r \tilde{b}_r^+$ and $N_0 = \sum_e b_e^-$. Source principals are subsets of negative charges, yielding

$$P_0 \leq \eta \sum_d D_d \leq \eta N_0. \quad (24)$$

Let C be the amount canceled when positive and negative events share actions. Net masses $P = P_0 - C$ and $N = N_0 - C$ satisfy

$$P - \eta N = P_0 - \eta N_0 - (1 - \eta)C \leq 0 \quad (25)$$

for $0 \leq \eta \leq 1$. For any $\rho > 0$ and $a, b \geq 0$, the clipped surrogate obeys

$$\phi_\epsilon(\rho, a) \leq (1 + \epsilon)a, \quad (26)$$

$$\phi_\epsilon(\rho, -b) \leq -(1 - \epsilon)b. \quad (27)$$

Averaging over each action’s tokens preserves these inequalities, yielding

$$\begin{aligned} J_i^{\text{loc}} &\leq (1 + \epsilon)P - (1 - \epsilon)N \\ &\leq [(1 + \epsilon)\eta - (1 - \epsilon)]N. \end{aligned} \quad (28)$$

This establishes Equation 17 under its stated condition. Within trajectories containing external bonuses, the bound applies to self-induced components with action-disjoint support or independently evaluated local surrogates. These conditions preserve component-level accounting under nonlinear clipping.

Shared policy parameters can increase error creation despite the scalar escrow bound. Consider an erroneous action with probability $p[\theta] = \sigma[\theta]$ and a subsequent repair with probability $q[\theta] = \sigma[H\theta]$.

Assign coefficients $-D$ to the source and ηD to its verified repair. At $\theta = 0$, the expected local score-gradient update is

$$g_{\text{loc}}[0] = -\frac{D}{4} + \frac{\eta DH}{8}. \quad (29)$$

For $H > 2/\eta$, the update increases error probability while satisfying the escrow constraint. The counterexample separates a bounded local surrogate from behavioral guarantees and motivates empirical assessment of error introduction, correction, and repeated repair cycles.

C.4 Observational Attribution Limits

A failed trajectory can support causal models with different intervention effects. For binary errors E_1, E_2 , both $F = E_1 \vee E_2$ and $F = E_1 \wedge E_2$ explain $E_1 = E_2 = F = 1$. Correcting E_1 preserves failure under the first model and resolves it under the second. The trace therefore leaves intervention-based responsibility underdetermined. ECHO uses the ownership convention in Equation 3 to allocate observed constraint impairment.

D Implementation Details

D.1 Training Data and Implementation

ECHO is initialized from Qwen3-4B and Qwen3-8B and trained through reinforcement learning. We use 92 tasks from the τ^2 -Bench training splits, comprising 66 retail and 26 airline tasks. A further 8 retail and 4 airline tasks form the 12-task development split. Both scales use 75 updates, 12 tasks per batch, eight rollouts per task, and a learning rate of 5×10^{-6} .

D.2 ECHO Configuration

Kimi-K2.6 serves as the fixed judge for retrospective proposals and independent prefix assessments. Each trajectory receives one proposal request and at most eight prefix assessments, with six proposal slots and two sentinel slots. We use $\kappa = 0.25$, $\alpha = 0.5$, $\lambda_- = \lambda_+ = 0.5$, $\eta = 0.5$, and clipping half-width $\epsilon = 0.2$. Outcome advantages use population-standard-deviation normalization with $\varepsilon = 10^{-6}$. Appendices A and B specify evidence validation and reliability gating.

E Optimization Details and Computational Cost

E.1 Update Sequence

Each update audits successful and failed trajectories and retains every group reward for outcome

normalization. Negative budgets establish source principals before eligible corrections undergo positive normalization and escrow. Aggregated action coefficients remain fixed during optimization of Equation 15; the judge operates during training. The coefficients in Section 4.1 satisfy the escrow condition $\eta \leq [1 - \epsilon]/[1 + \epsilon]$.

Stored behavior probabilities correspond to the rollout distribution, including its temperature and token support. The sampling convention uses temperature one with full token support, and distributed accumulation averages over contributing trajectories across workers.

E.2 Audit Cost

Auditing a trajectory uses one proposal request and at most K independent prefix requests. For input lengths n and output lengths m , the token cost is

$$C_i^{\text{judge}} = n_i^{\text{full}} + m_i^{\text{full}} + \sum_{t \in S_i} [n_{i,t}^{\text{pre}} + m_{i,t}^{\text{pre}}] + C_i^{\text{retry}}. \quad (30)$$

Requirement-inventory compilation is amortized per task and accounted for separately. Uncached prefix input scales up to $O(Kn_i)$; exact-prefix caching can reduce repeated computation, while batching preserves request count. Inference cost also depends on context length and attention implementation.

E.3 Realized Local Coefficients

Away from clipping boundaries, an active local token contributes $\delta_{i,t}\rho_{i,k}/B_{i,t}$ to its score-gradient coefficient. Define

$$\chi_{i,t,k} = \mathbf{1}\{\delta_{i,t} > 0, \rho_{i,k} \leq 1 + \epsilon\} + \mathbf{1}\{\delta_{i,t} < 0, \rho_{i,k} \geq 1 - \epsilon\}, \quad (31)$$

$$D_{i,t}^{\text{eff}} = \frac{\delta_{i,t}}{B_{i,t}} \sum_k m_{i,t,k} \rho_{i,k} \chi_{i,t,k}. \quad (32)$$

Active negative-advantage ratios can exceed the upper clipping threshold, so realized coefficient magnitudes can exceed their on-policy masses. Realized coefficients and sign-specific clipping fractions characterize how supervision changes through aggregation and clipping. Gradient norms and local-to-outcome alignment additionally distinguish update magnitude from directional agreement.

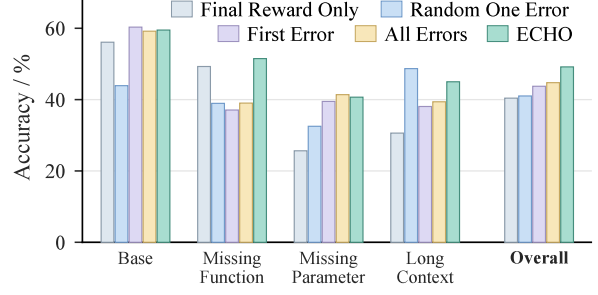


Figure 6: Reward ablation on BFCL-v3 Multi-Turn at 8B. Overall is the mean across Base, Missing Function, Missing Parameter, and Long Context.

F Reward Ablation on Multi-Turn Function Calling

The BFCL-v3 Multi-Turn ablation tests supervision under distinct information constraints, as shown in Figure 6. ECHO exceeded Final Reward Only by 8.76 percentage points and All Errors, the strongest ablation, by 4.42 points overall. Its advantage over All Errors was concentrated in Missing Function, where all three error-selection heuristics fell below terminal supervision. ECHO improved this subset while retaining gains over Final Reward Only in Missing Parameter and Long Context.

Subset leaders varied: First Error led Base, All Errors led Missing Parameter, and Random One Error led Long Context. The Long Context advantage of Random One Error accompanied a substantial Base deficit, illustrating the uneven effects of unstructured selection. The strongest heuristic also changed from First Error on τ^2 -Bench to All Errors on BFCL-v3 Multi-Turn. ECHO’s aggregate lead across both benchmarks supports its balance across decision conditions, alongside the observed advantages of simpler rules on individual subsets.